



INTRODUCTION TO PRINCIPAL COMPONENT ANALYSIS (PCA)

INT301 Bio-computation, Week 10, 2025



Preliminary Knowledge

Eigenvalues and Eigenvectors

如果 v 是一个非零向量, 且 λ 是一个数

- If v is a nonzero vector and λ is a number such that

$$\underline{Av = \lambda v}$$

then v is said to be an *eigenvector* of A with

eigenvalue λ . v 被称为矩阵 A 的特征向量, λ 被称为对应的特征值.

Example

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \times \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 3 \times \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

A

V (eigenvectors)

λ (eigenvalues)

当矩阵 A 乘特征向量 v 时, 结果向量 Av 只是 v 的一个缩放 (伸长或缩短), 方向保持不变。除非 $\lambda < 0$, 缩放的因子就是特征值 λ 。

Preliminary Knowledge

Eigenvalues and Eigenvectors

□ Eigenvectors (for a square $m \times m$ matrix S)

对于一个 $m \times m$ 的方阵 S , 特征值 λ 的求解基于以下方程:

$$Sv = \lambda v \Leftrightarrow (S - \lambda I)v = 0$$

eigenvector

$$v \in \mathbb{R}^m \neq 0$$

$$Sv = \lambda v$$

eigenvalue

$$\lambda \in \mathbb{R}$$

Example

$$\begin{pmatrix} 6 & -2 \\ 4 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

要使这个方程有非0解 v ,

矩阵 $(S - \lambda I)$ 必须是奇异的, 即它的行列式必须为0.

□ How many eigenvalues are there at most?

$$Sv = \lambda v \iff (S - \lambda I)v = 0$$

only has a non-zero solution if $|S - \lambda I| = 0$

with **at most m distinct λ values**

对于一个 $m \times m$ 矩阵, 通过求解这个 m 阶多项式, 最多可以得到 m 个不同的特征值

PCA中通常处理“协方差矩阵”

Preliminary Knowledge

Eigenvalues and Eigenvectors

正交性(orthogonal): 对于对称矩阵 S , 对应于不同特征值 ($\lambda_1 \neq \lambda_2$) 的特征向量是正交的。
For symmetric matrices, eigenvectors for distinct eigenvalues are **orthogonal** 前提是对称矩阵 S ($S=S^T$)

$$Sv_{\{1,2\}} = \lambda_{\{1,2\}} v_{\{1,2\}}, \text{ and } \lambda_1 \neq \lambda_2 \Rightarrow v_1 \bullet v_2 = 0$$

All eigenvalues of a real symmetric matrix are **real**.

一个实数对称矩阵的所有特征值都是实数

All eigenvalues of a positive semidefinite matrix are **non-negative**

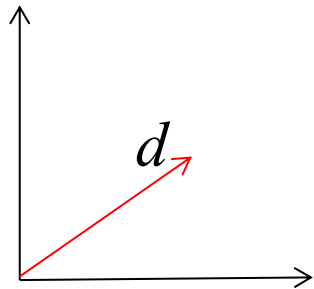
一个半正定矩阵的所有特征值都是非负的

$$\forall w \in \mathbb{R}^n, w^T S w \geq 0, \text{ then if } S v = \lambda v \Rightarrow \lambda \geq 0$$

positive semidefinite matrix: 对于任何向量 $w \in \mathbb{R}^n$, 都有 $w^T S w \geq 0$. 若 S 是半正定, 则特征值 $\lambda \geq 0$.

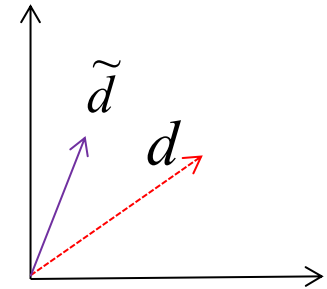
Preliminary Knowledge

Eigenvalues and Eigenvectors



$$S = S^T, \quad S v_i = \lambda_i v_i, \quad (\lambda_1 = 2, \lambda_2 = 0.5)$$

$$Sd = \begin{bmatrix} S(1,1) & S(1,2) \\ S(2,1) & S(2,2) \end{bmatrix} \begin{bmatrix} d(1) \\ d(2) \end{bmatrix} = \begin{bmatrix} S(1,1)d(1) + S(1,2)d(2) \\ S(2,1)d(1) + S(2,2)d(2) \end{bmatrix} = \tilde{d}$$



投影 (projection): 向量 d 可以分解为沿着特征向量 v_1 和 v_2 方向的分量 d_1 和 d_2 .

缩放 (scaling): 当 S 位于 d 时, 其效果相当于将 d 投影到每个特征向量上, 然后用对应的特征值 λ_i 来缩放这些分量。

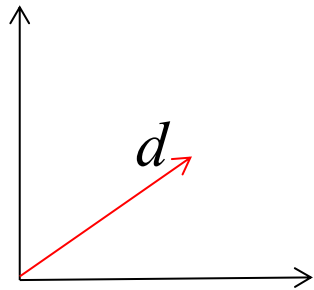
在下图中, 分量 d_1 被 λ_1 缩放成 $\lambda_1 d_1$, 分量 d_2 被 λ_2 缩放成 $\lambda_2 d_2$.

向量加法 (vector addition): 缩放后的分量相加, 就得到了变换后的向量 $\tilde{d}$

$$Sd = \tilde{d}$$

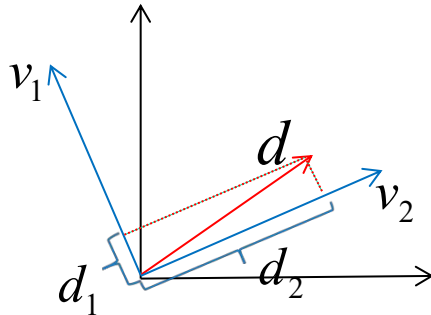
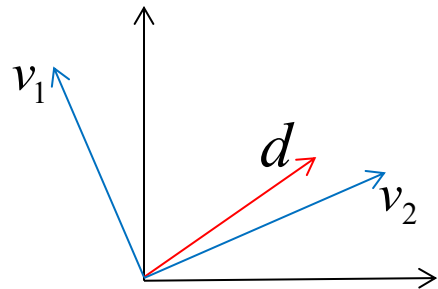
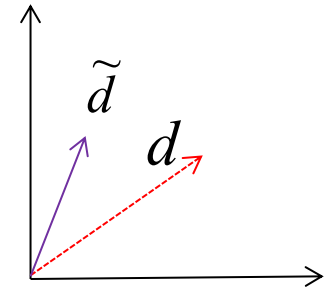
如果 v_1, v_2 是标准化的特征向量, 那么 $d = d_1 v_1 + d_2 v_2$, 且 $\tilde{d} = Sd = S(d_1 v_1 + d_2 v_2) = d_1 (Sv_1) + d_2 (Sv_2) = d_1 (\lambda_1 v_1) + d_2 (\lambda_2 v_2)$.

Preliminary Knowledge Eigenvalues and Eigenvectors

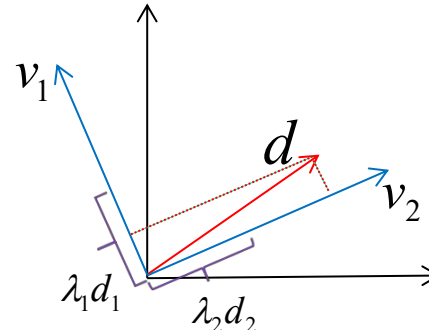


$$S = S^T, \quad S v_i = \lambda_i v_i, \quad (\lambda_1 = 2, \lambda_2 = 0.5)$$

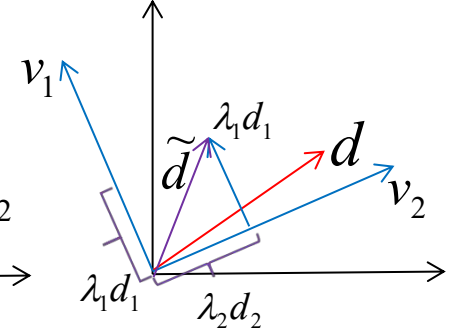
$$S d = \begin{bmatrix} S(1,1), S(1,2) \\ S(2,1), S(2,2) \end{bmatrix} \begin{bmatrix} d(1) \\ d(2) \end{bmatrix} = \begin{bmatrix} S(1,1)d(1) + S(1,2)d(2) \\ S(2,1)d(1) + S(2,2)d(2) \end{bmatrix} = \tilde{d}$$



Projection on the **eigenvectors**



Scaling with **eigenvalues**



Vector addition

Preliminary Knowledge

Eigenvalues and Eigenvectors

■ Let 矩阵 S .

$$S = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

实数对称矩阵.

Real, symmetric.

■ Then 求解特征值: 计算 $|Sv - \lambda I| = 0$
$$S - \lambda I = \begin{bmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{bmatrix} \Rightarrow (2 - \lambda)^2 - 1 = 0.$$

■ The eigenvalues are 1 and 3 (nonnegative, real). $\lambda = 1$ 和 3 .

■ The eigenvectors are orthogonal (and real):

求特征向量 v : 将特征值 $\lambda = 1$ 与 $\lambda = 3$ 分别代入 $(S - \lambda I)v = 0$

中求解对应的特征向量 $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$ 和 $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$

• result: $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$ 和 $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$

这两个特征向量是正交的, 符合对称矩阵的性质.

Substitute these values
and solve for
eigenvectors.

Diagonal Decomposition

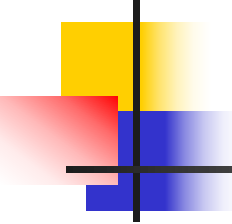
- Let $S \in \mathbb{R}^{m \times m}$ be a **square** matrix with m **linearly independent eigenvectors**
- **Theorem:** Exists an **eigen decomposition**

$$S = U \Lambda U^{-1}$$

diagonal

Unique
for
distinct
eigen-
values

- (matrix diagonalization theorem)
- Columns of U are **eigenvectors** of S
- Diagonal elements of Λ are **eigenvalues** of S
 $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_m), \quad \lambda_i \geq \lambda_{i+1}$



Diagonal Decomposition

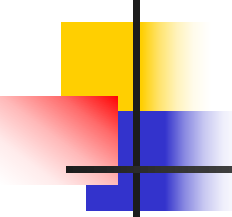
Write **U** with the eigenvectors as columns: $U = \begin{bmatrix} v_1 & \dots & v_m \end{bmatrix}$

Then, **SU** can be written

$$SU = S \begin{bmatrix} v_1 & \dots & v_m \end{bmatrix} = \begin{bmatrix} \lambda_1 v_1 & \dots & \lambda_m v_m \end{bmatrix} = \begin{bmatrix} v_1 & \dots & v_m \end{bmatrix} \begin{bmatrix} \lambda_1 & & \\ & \dots & \\ & & \lambda_m \end{bmatrix}$$

Thus **$SU = U\Lambda$** , or **$U^{-1}SU = \Lambda$**

Therefore **$S = U\Lambda U^{-1}$** .



Diagonal Decomposition

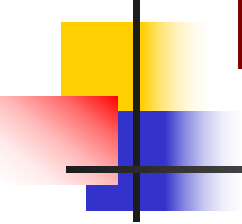
Recall $S = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}; \lambda_1 = 1, \lambda_2 = 3.$

The eigenvectors $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ form $U = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$

Inverting U , we have

$$U^{-1} = \begin{bmatrix} 1/2 & -1/2 \\ 1/2 & 1/2 \end{bmatrix}$$

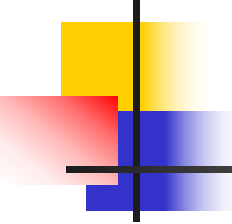

$$\text{Then, } \mathbf{S} = \mathbf{U} \mathbf{\Lambda} \mathbf{U}^{-1} = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1/2 & -1/2 \\ 1/2 & 1/2 \end{bmatrix} \quad 10$$



Diagonal Decomposition

Let's divide $\mathbf{U}$ (and multiply $\mathbf{U}^{-1}$) by $\sqrt{2}$

$$\text{Then, } \mathbf{S} = \underbrace{\begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}}_{\mathbf{Q}} \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}}_{\mathbf{\Lambda}} \underbrace{\begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}}_{(\mathbf{Q}^{-1} = \mathbf{Q}^T)}$$



Symmetric Diagonal Decomposition

- **Theorem:** If $S \in \mathbb{R}^{m \times m}$ is a **symmetric** matrix, there exists an **eigen decomposition**, where Q is **orthogonal**:

$$S = Q\Lambda Q^T$$

- $Q^{-1} = Q^T$
- Columns of Q are normalized eigenvectors
- Columns are orthogonal.
- (everything is real)

Singular Value Decomposition

For an $m \times n$ matrix $\mathbf{A}$ of rank r , there exists a factorization (Singular Value Decomposition = **SVD**) as follows:

$$\mathbf{A} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^T$$

$m \times m$ $m \times n, \text{ nonnegative real}$ $n \times n$

The columns of $\mathbf{U}$ are orthogonal eigenvectors of $\mathbf{A}\mathbf{A}^T$.

The columns of $\mathbf{V}$ are orthogonal eigenvectors of $\mathbf{A}^T\mathbf{A}$.

Eigenvalues $\lambda_1 \dots \lambda_r$ of $\mathbf{A}\mathbf{A}^T$ are the eigenvalues of $\mathbf{A}^T\mathbf{A}$.

$$\sigma_i = \sqrt{\lambda_i}$$
$$\mathbf{\Sigma} = \text{diag}(\sigma_1 \dots \sigma_r)$$

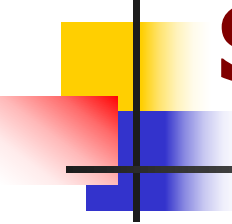
Singular values.

Singular Value Decomposition

- Illustration of SVD dimensions and

$$\underbrace{\begin{bmatrix} * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \end{bmatrix}}_A = \underbrace{\begin{bmatrix} * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \end{bmatrix}}_U \underbrace{\begin{bmatrix} \bullet & & & & \\ & \bullet & & & \\ & & \bullet & & \\ & & & \bullet & \\ & & & & \bullet \end{bmatrix}}_{\Sigma} \underbrace{\begin{bmatrix} * & * & * \\ * & * & * \\ * & * & * \end{bmatrix}}_{V^T}$$

$$\underbrace{\begin{bmatrix} * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \end{bmatrix}}_A = \underbrace{\begin{bmatrix} * & * & * \\ * & * & * \\ * & * & * \end{bmatrix}}_U \underbrace{\begin{bmatrix} \bullet & & & & \\ & \bullet & & & \\ & & \bullet & & \\ & & & \bullet & \\ & & & & \bullet \end{bmatrix}}_{\Sigma} \underbrace{\begin{bmatrix} * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \end{bmatrix}}_{V^T}$$



Singular Value Decomposition

$$\text{Let } A = \begin{bmatrix} 1 & -1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Thus $m=3$, $n=2$. Its SVD is

$$\begin{bmatrix} 0 & 2/\sqrt{6} & 1/\sqrt{3} \\ 1/\sqrt{2} & -1/\sqrt{6} & 1/\sqrt{3} \\ 1/\sqrt{2} & 1/\sqrt{6} & -1/\sqrt{3} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & \sqrt{3} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}$$

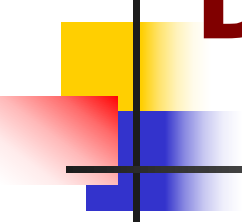
Dimensionality Reduction

- One approach to deal with high dimensional data is by reducing their dimensionality.
- Project high dimensional data onto a lower dimensional **subspace** using linear or non-linear transformations.

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ x_N \end{bmatrix} \xrightarrow{\text{Reduce dimensionality}} Y = \begin{bmatrix} y_1 \\ y_2 \\ \dots \\ y_k \end{bmatrix} \quad (K \ll N)$$

- Linear transformations are simple to compute

$$\begin{array}{ccccc} Y & = & U & X & (b_i = u_i^t a_i) \\ \nearrow & & \nearrow & \nwarrow & \\ k \times 1 & & k \times d & d \times 1 & (k \ll d) \end{array}$$



Dimensionality Reduction

- **Find a basis in a low dimensional sub-space:**

- Approximate vectors by projecting them in a low dimensional sub-space:

(1) Original space representation:

$$x = a_1 v_1 + a_2 v_2 + \dots + a_N v_N$$

where $v_1, v_2, \dots, v_n$ is a base in the original N-dimensional space

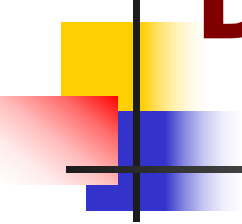
$$\begin{bmatrix} a_1 \\ a_2 \\ \dots \\ a_N \end{bmatrix}$$

(2) Lower-dimensional sub-space representation:

$$\hat{x} = b_1 u_1 + b_2 u_2 + \dots + b_K u_K$$

where $u_1, u_2, \dots, u_K$ is a base in the K -dimensional sub-space ($K < N$)

$$\begin{bmatrix} b_1 \\ b_2 \\ \dots \\ b_k \end{bmatrix}$$



Dimensionality Reduction

- If $K=N$, then $\hat{x} = x$
- Example ($K=N$):

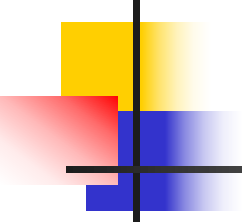
$$v_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, v_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, v_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad (\text{standard basis})$$

$$x_v = \begin{bmatrix} 3 \\ 3 \\ 3 \end{bmatrix} = 3v_1 + 3v_2 + 3v_3$$

$$u_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, u_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, u_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad (\text{some other basis})$$

$$x_u = \begin{bmatrix} 3 \\ 3 \\ 3 \end{bmatrix} = 0u_1 + 0u_2 + 3u_3$$

thus, $x_v = x_u$



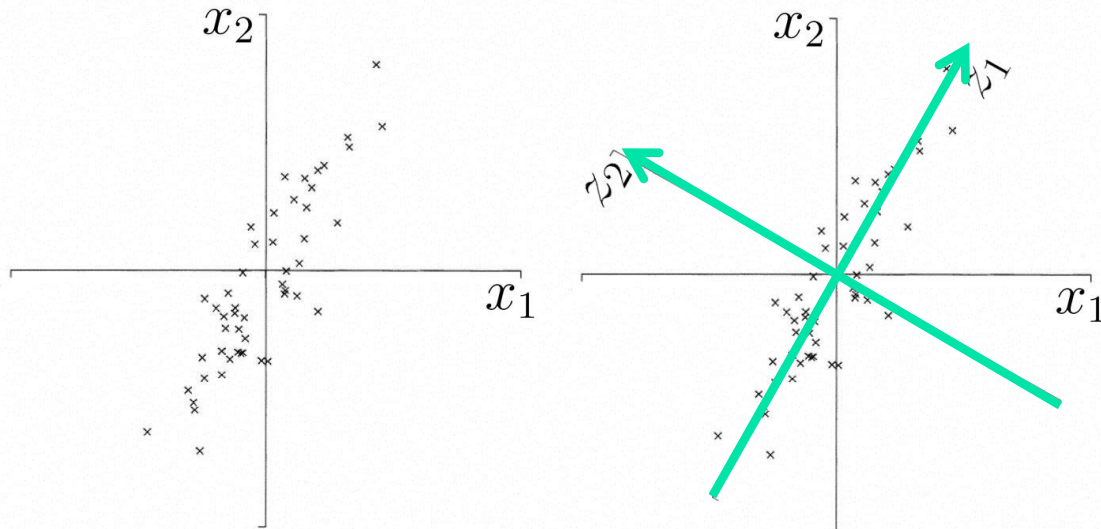
Principal Component Analysis (PCA)

- Each dimensionality reduction technique finds an *appropriate transformation* by satisfying certain criteria (e.g., *information loss*, *data discrimination*, etc.)
- The goal of PCA is to reduce the dimensionality of the data while *retaining as much as possible of the variation present in the dataset*.

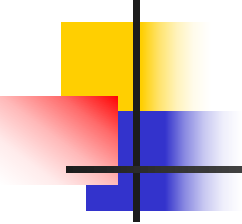
Principal Component Analysis (PCA)

■ Motivation

- Find bases which has high variance in data
- Encode data with small number of bases with low MSE



- First PC is direction of maximum variance
- Subsequent PCs are orthogonal to 1st PC and describe maximum residual variance



Principal Component Analysis (PCA)

Assume that $E[\mathbf{x}] = \mathbf{0}$ $a = \mathbf{x}^T \mathbf{q} = \mathbf{q}^T \mathbf{x}$ $\|\mathbf{q}\| = (\mathbf{q}^T \mathbf{q})^{1/2} = 1$

→ $\sigma^2 = E[a^2] - E[a]^2 = E[a^2]$
 $= E[(\mathbf{q}^T \mathbf{x})(\mathbf{x}^T \mathbf{q})] = \mathbf{q}^T E[\mathbf{x}\mathbf{x}^T] \mathbf{q} = \mathbf{q}^T \mathbf{R} \mathbf{q}$

Find $\mathbf{q}$'s maximizing variance!!

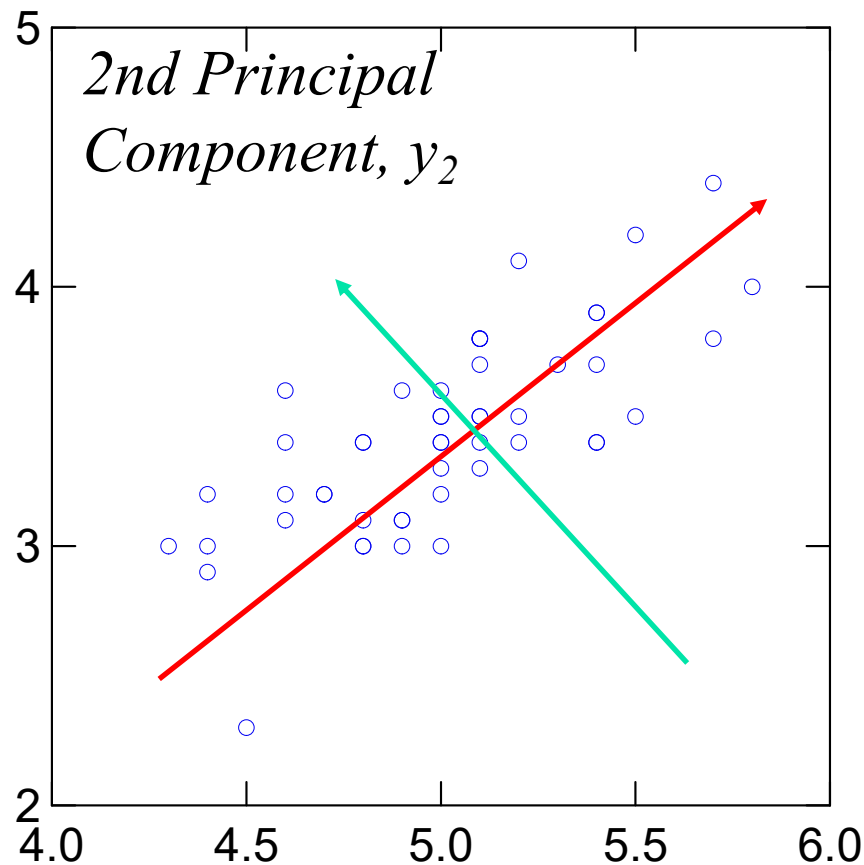
It can be shown that variance is maximized when $\mathbf{q}$ is the principal component of $\mathbf{R}$.

Principal component $\mathbf{q}$ can be obtained by **eigenvector decomposition**:

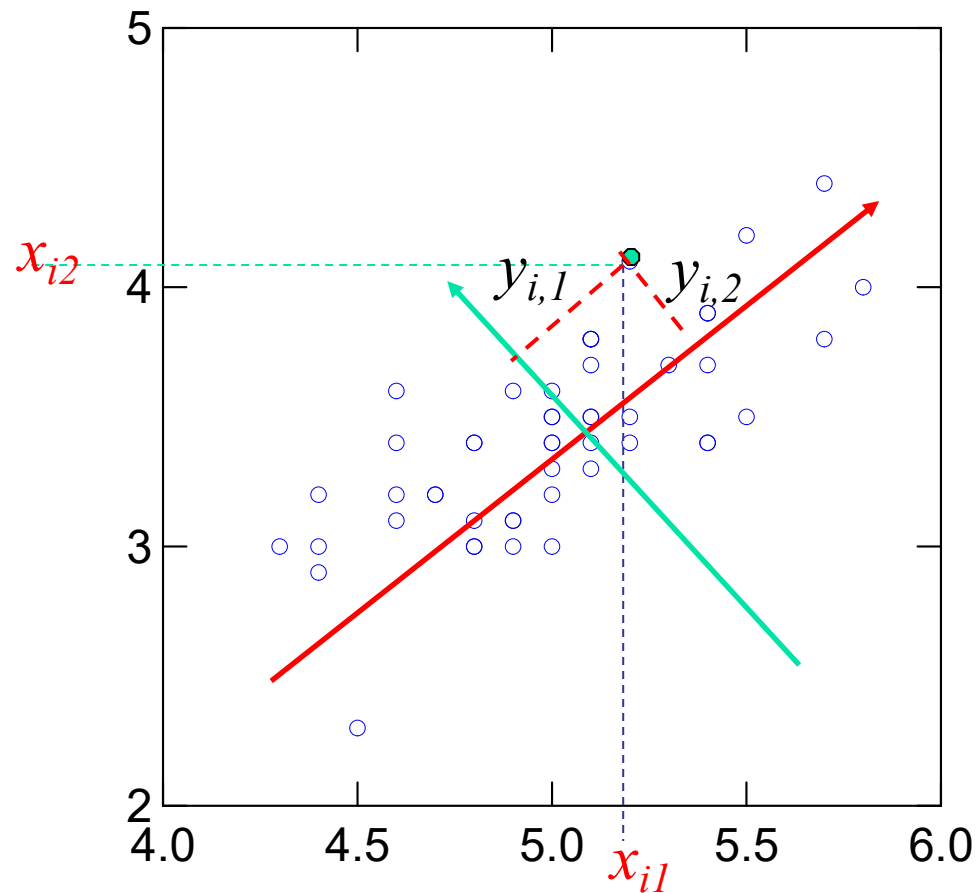
$$\mathbf{R} = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^T, \quad \mathbf{Q} = [\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_j, \dots, \mathbf{q}_m], \quad \mathbf{\Lambda} = \text{diag}[\lambda_1, \lambda_2, \dots, \lambda_j, \dots, \lambda_m]$$

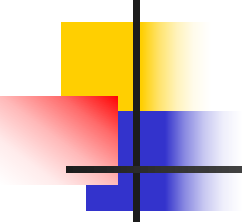
$$\Leftrightarrow \mathbf{R} \mathbf{q}_j = \lambda_j \mathbf{q}_j \quad j = 1, 2, \dots, m \quad \rightarrow \quad \mathbf{R} \mathbf{q} = \lambda \mathbf{q}$$

Principal Component Analysis (PCA)



Principal Component Analysis (PCA)

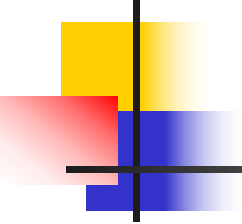




Principal Component Analysis (PCA)

- Advantage

- Reduce the dimension of the original data
 - reduce time consumption in the training process, and improve efficiency
- Discard some information of the original data
 - if this information is noise



Principal Component Analysis (PCA)

- Limitation

- Discard some information of the original data
 - if the discarded information is important, it is not appropriate to apply PCA
- The meaning of the principal component
 - PC or basis may not be interpretable
- Linear model of PCA
 - not suitable for nonlinear problem
- Assume first PC has higher importance

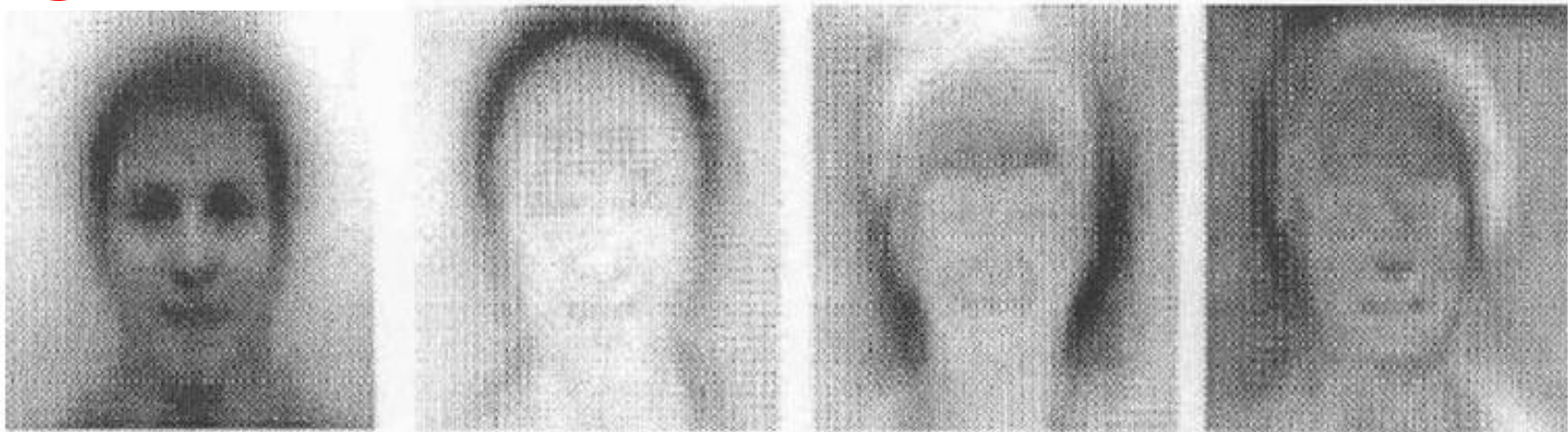
Case study: Eigenface

Face image: high-dimensional vector

face image = linear combination of eigenvectors

The eigenvectors can be viewed as images.

Eigenfaces:



b_1

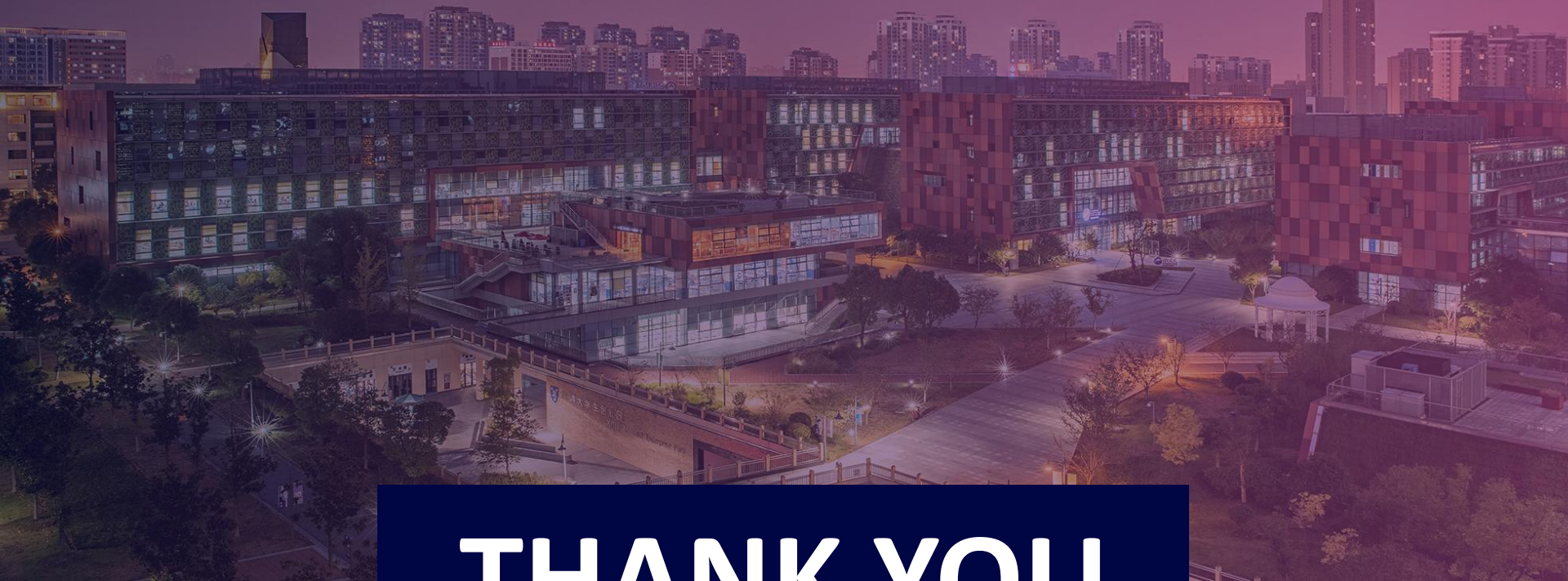
b_2

b_3

b_4



Original
face image



THANK YOU



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